

26 A **1st circuit:** Let supply be V_S

$$\frac{V_Q}{V_P} = \frac{N_Q}{N_P} = \frac{20}{1}$$

$$V_Q = 20 V_P = 20 V_S$$

$$\text{current through } R = \frac{V_Q}{R}$$

$$2.0 = \frac{20V_S}{R}$$

$$\frac{V_S}{R} = 0.10$$

2nd circuit:

$$\frac{V_P'}{V_Q'} = \frac{N_P}{N_Q} = \frac{1}{20}$$

$$V_P' = \frac{1}{20} V_Q' = 0.050 V_S$$

$$\text{current through } R = \frac{V_P'}{R} = \frac{0.050 V_S}{R} = 0.050 \times 0.10 = 0.0050 \text{ A}$$

